Re=1000 Audit: the base regime on sequences no model ever trained on

Baseline inference, the dials alone and coupled, the depth ladder, and the fine-tuned variants. Fixed K3x86 unless stated. Held-out sequences 34-39 of the 40-seed file (base pretrain 0-31, val 32-35 with the fine-tunes on 32-33, test 36-39); 20 frames each. 2026-08-25.

Provenance note

The earlier cross-regime grid evaluated Re=1000 on sequences 4-19, which are base-TRAINING sequences; this audit supersedes those numbers. The measured memorization inflation was small (base K3 unguided: 0.48/0.92 there vs 0.46/0.91 here), so no earlier conclusion flips, but every Re=1000-specific number in the write-up should come from this audit.

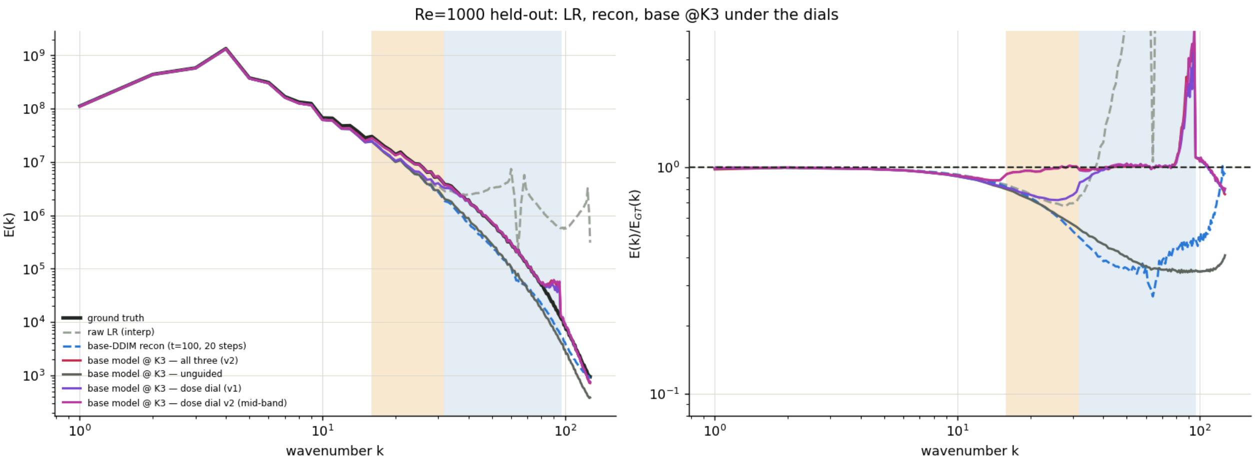
1 · The observation and the recon

row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place	MSE	resid
raw LR	0.99	0.95	0.76	2.37	20.62	0.669	1.28	370.58
base-DDIM recon	0.99	0.95	0.71	0.41	0.38	0.745	0.37	0.82

The LR's above-Nyquist "retention" (2.4x / 21x) is aliased excess, not content. The recon strips it, keeps 0.41 of the fine band, places at 0.745, and has the best MSE of any row - the well-placed conditional mean.

2 · The base model under the dials, alone and coupled

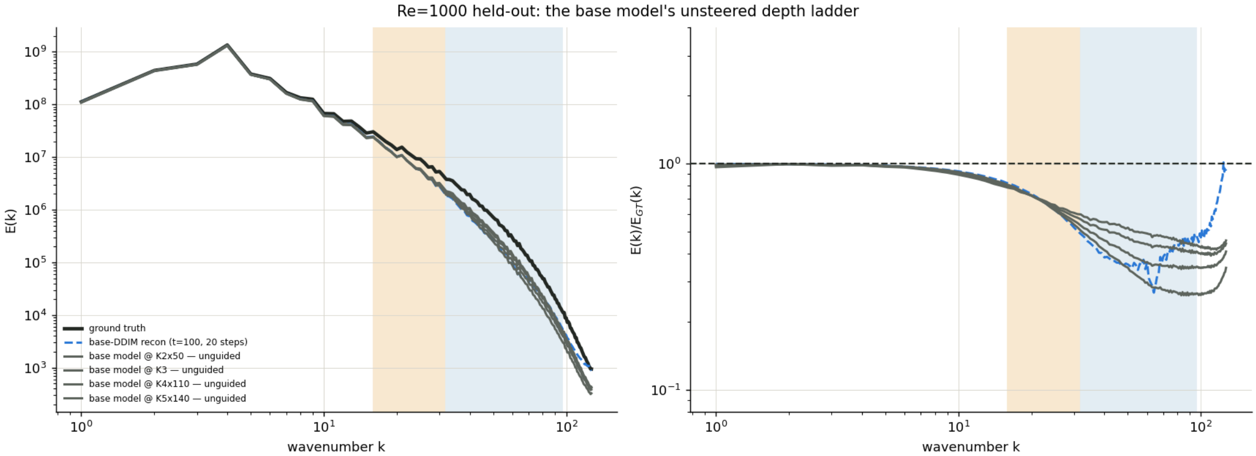
row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place	MSE	resid
base K3 all three v1	0.99	0.94	0.76	0.95	1.16	0.741	0.43	1.51
base K3 all three v2	0.99	0.95	0.97	0.99	1.18	0.732	0.46	1.62
base K3 unguided	0.99	0.94	0.71	0.46	0.35	0.908	0.40	0.62
base K3 placement	0.99	0.94	0.70	0.45	0.35	0.855	0.41	0.65
base K3 residual	0.99	0.94	0.71	0.46	0.34	0.907	0.40	0.49
base K3 dose v1	0.99	0.94	0.76	0.95	1.13	0.753	0.43	1.62
base K3 dose v2	0.99	0.95	0.97	0.99	1.18	0.750	0.46	1.82



In-regime every dose-adding dial takes the base to ~1.0 retention but pushes its residual above the true flow's level (1.6-1.9x) and costs 0.15 placement; the residual dial polishes; the placement dial HURTS the base here (0.91 to 0.86 - the LR-map ceiling); v2 closes the mid band (0.71 to 0.96).

3 · Inference depth: the base's unsteered ladder

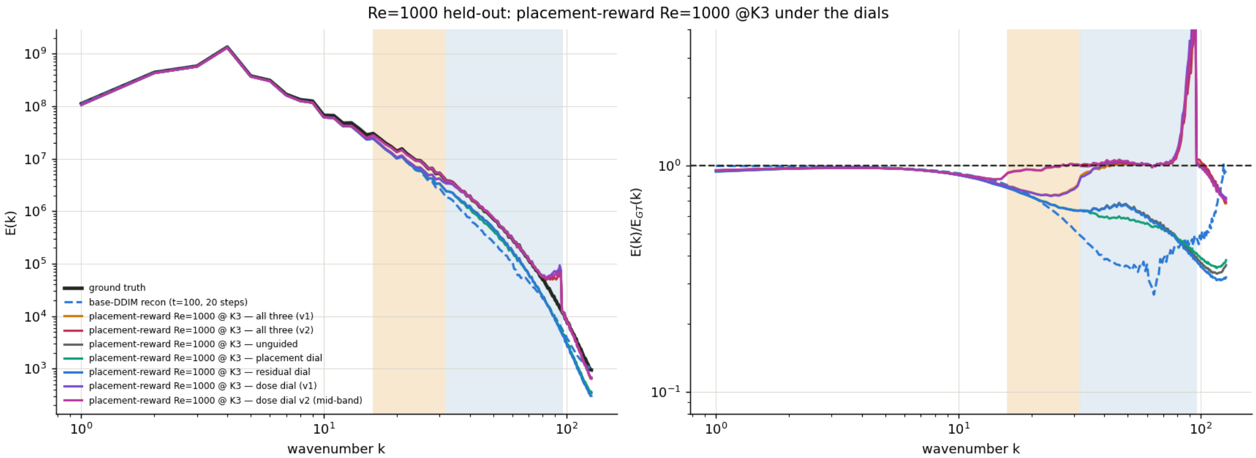
row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place	MSE	resid
base K2x50 unguided	0.99	0.95	0.71	0.42	0.28	0.900	0.37	0.57
base K3 unguided	0.99	0.94	0.71	0.46	0.35	0.908	0.40	0.62
base K4x110 unguided	0.99	0.94	0.71	0.51	0.43	0.901	0.48	0.66
base K5x140 unguided	0.98	0.93	0.71	0.55	0.47	0.883	0.59	0.67



K2 to K5: retention 0.42 to 0.55 only, placement slipping, MSE rising - every rung sags to 0.25-0.45 through the fine band. Depth is the wrong lever in-regime.

4 · placement-reward Re=1000 (pr1k-549)

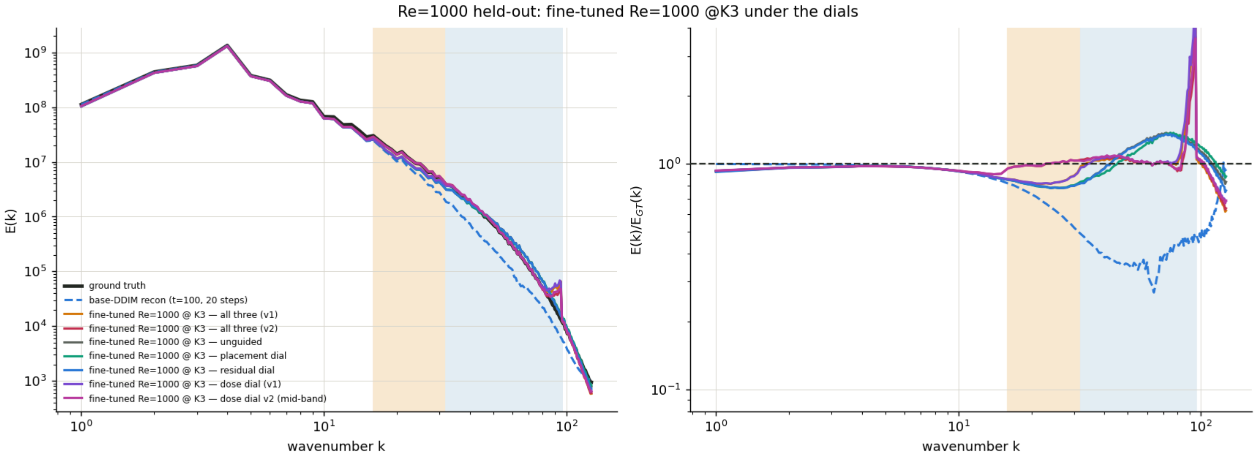
row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place	MSE	resid
placement-reward Re=1000 K3 all three v1	0.98	0.94	0.77	0.97	1.23	0.729	0.48	1.87
placement-reward Re=1000 K3 all three v2	0.98	0.95	0.96	1.01	1.20	0.714	0.52	1.90
placement-reward Re=1000 K3 unguided	0.98	0.94	0.72	0.64	0.52	0.779	0.46	0.98
placement-reward Re=1000 K3 placement	0.98	0.94	0.72	0.60	0.51	0.840	0.45	1.03
placement-reward Re=1000 K3 residual	0.97	0.94	0.72	0.64	0.51	0.778	0.46	0.79
placement-reward Re=1000 K3 dose v1	0.98	0.94	0.77	0.98	1.28	0.676	0.49	2.12
placement-reward Re=1000 K3 dose v2	0.98	0.95	0.96	1.02	1.24	0.681	0.52	2.04



5 · ft Re=1000 (r1k-449)

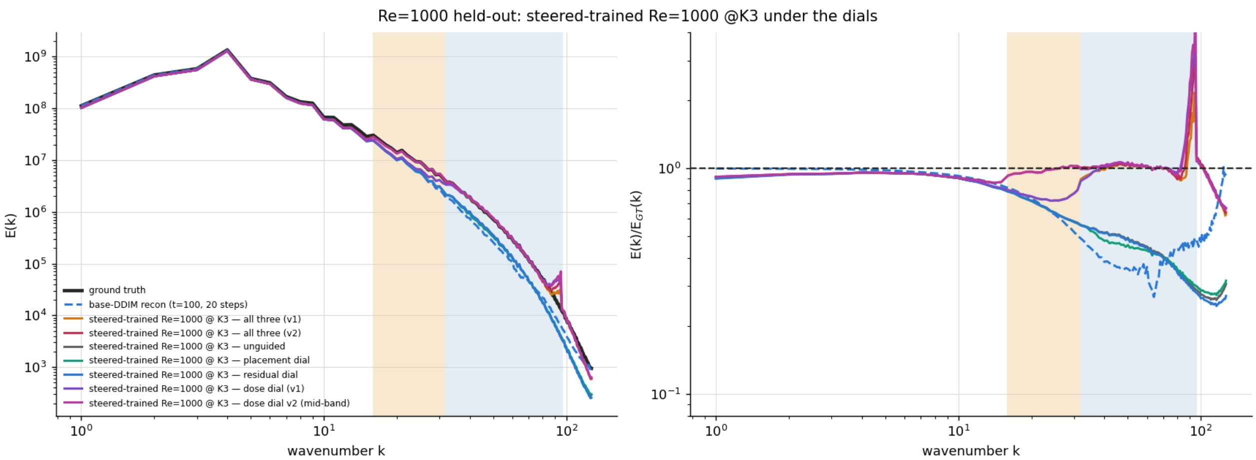
row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place	MSE	resid
ft Re=1000 K3 all three v1	0.97	0.95	0.84	1.02	1.15	0.739	0.45	1.33
ft Re=1000 K3 all three v2	0.97	0.95	0.99	1.05	1.08	0.726	0.48	1.20
ft Re=1000 K3 unguided	0.97	0.95	0.81	0.97	1.33	0.863	0.43	0.79

ft Re=1000 K3 placement	0.97	0.95	0.81	0.94	1.32	0.863	0.43	0.86
ft Re=1000 K3 residual	0.97	0.95	0.81	0.96	1.31	0.862	0.43	0.64
ft Re=1000 K3 dose v1	0.97	0.95	0.84	1.03	1.18	0.740	0.45	1.47
ft Re=1000 K3 dose v2	0.97	0.95	0.99	1.06	1.09	0.736	0.48	1.32



6 · steered-trained Re=1000 (st1k-599)

row	[1,5)	[5,16)	[16,32)	[32,64)	[64,96)	place	MSE	resid
steered-trained Re=1000 K3 all three v1	0.95	0.93	0.76	0.98	1.01	0.727	0.48	1.68
steered-trained Re=1000 K3 all three v2	0.95	0.93	0.97	1.02	1.05	0.717	0.51	1.88
steered-trained Re=1000 K3 unguided	0.95	0.93	0.71	0.52	0.39	0.874	0.45	1.20
steered-trained Re=1000 K3 placement	0.95	0.93	0.70	0.50	0.39	0.847	0.45	1.23
steered-trained Re=1000 K3 residual	0.95	0.93	0.70	0.52	0.38	0.873	0.45	1.00
steered-trained Re=1000 K3 dose v1	0.95	0.93	0.76	0.98	1.10	0.734	0.48	2.13
steered-trained Re=1000 K3 dose v2	0.95	0.93	0.97	1.02	1.12	0.729	0.51	2.29



Fine-tune vs steering, in-regime: the r1k fine-tune reaches the same retention as base+dial (0.99 vs 1.00) while placing the energy (+0.11) and keeping the flow physical (residual 0.79 below the true level vs 1.82 above it). Putting the dial on top of a well-trained in-regime model is harmful (placement -0.13, residual x1.7). Doctrine: fine-tune at home, steer away from home.

Per-sample validation

row	ret paired med [p10,p90]	place/frame med [p10,p90]	MSE med	frames in [0.8,1.2]
placement-reward Re=1000 K3 all three v1	1.03 [0.64,2.77]	0.834 [0.740,0.896]	0.46	41%
placement-reward Re=1000 K3 all three v2	1.06 [0.64,3.03]	0.840 [0.761,0.898]	0.49	38%
placement-reward Re=1000 K3 unguided	0.63 [0.53,0.75]	0.796 [0.637,0.907]	0.45	6%
placement-reward Re=1000 K3 placement	0.59 [0.49,0.70]	0.817 [0.708,0.878]	0.44	2%
placement-reward Re=1000 K3 residual	0.63 [0.53,0.75]	0.796 [0.637,0.907]	0.45	5%
placement-reward Re=1000 K3 dose v1	1.03 [0.65,2.77]	0.796 [0.642,0.909]	0.47	40%
placement-reward Re=1000 K3 dose v2	1.08 [0.65,3.00]	0.813 [0.669,0.915]	0.50	38%